ABDULLAH AL MARUF

SOFTWARE ENGINEER

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Austin, TX, USA

PROFILE

Passionate Software Engineer with **7 years** of experience developing **Kubernetes** tools in **Go** and building cloud-native infrastructure using **Infrastructure as Code (IaC)** with **Terraform**, Helm, and ArgoCD across **AWS** and **Azure**. Skilled in backend development with **Java** and algorithmic problem solving in **C++**. Led the KubeDB project at AppsCode and contributed to open-source tools.

EXPERIENCE

Software Engineer II

Oct, 2022 - Present

ZEISS Medical Technology

Temple, TX, USA

- Collaborated with cross-functional teams to analyze and design high-level system architecture.
- Migrated software and infrastructure from **AWS ECS** to Kubernetes on **Azure AKS**, achieving **~3x faster** system performance.
- Designed and implemented infrastructure as code (IaC) with Terraform, **Helm**, and **ArgoCD**, with monitoring and alerting via **Prometheus** and Grafana.
- Refactored **backend** services in **Nodejs** to seamlessly integrate with AWS and Azure resources, unifying the codebase for cloud-agnostic and scalable development.

Intern

Summer 2021, 2022

Red Hat

Brno, CZ (Remote)

- Led a small team of students to conduct research on technical debt and microservice anti-patterns.
- Performed literature mapping study on technical debt and code smells.
- Technologies: Java, Spring Boot, Git.

DevOps Engineer

Feb, 2020 - Dec, 2020

Grameen Digital Health

Dhaka, Bangladesh

- Designed and migrated **scalable** cloud infrastructure to Kubernetes, ensuring high availability and resilience.
- Managed hybrid cloud infrastructure consisting of AWS and on-premises **OpenStack**, implementing automation pipelines for deployment and scaling.
- Optimized **CI/CD** workflows using Terraform, **Ansible**, **Jenkins**, and Docker, improving deployment consistency and reducing manual effort.
- Improved service **auto-scaling**, reducing infrastructure costs by **~25%**.
- Implemented monitoring, logging, and alerting pipelines using Prometheus and Grafana, enhancing system reliability and observability.

Software Engineer

May, 2017 - Jan, 2020

AppsCode Ltd.

Dhaka, Bangladesh

- Developed Kubernetes controller tools in **Go** and Docker to extend Kubernetes functionality.
- Designed and implemented KubeDB, a Kubernetes-native, cloud-agnostic database management tool built to operate consistently across GCP, AWS, and Azure Kubernetes clusters.
- Led a team of 5 to develop core KubeDB features, improving reliability and increasing returning users by **~200%**.
- Enhanced **MongoDB**, **Elasticsearch**, and **PostgreSQL** clustering support.

EDUCATION

Baylor University

M.S. in Computer Science

Chittagong University of Engineering & Technology

B.Sc. in Computer Science & Engineering

PROUD OF

Competitive Programming

- Solved **1100+** problems on different Online Judges.
- **Divisional Champion** of ACM ICPC Dhaka Regional '14 & '15.

Open Source Participation

User, follower, and contributor.

Research Impact

- **600+** citations across peer-reviewed publications
- Focus: microservices architecture, observability, and system design

KubeDB's growth

From very few to thousands of daily users.

Leadership

Served as President of Cuet Computer Club. Also acted as team lead of KubeDB.

FIND ME ONLINE

- https://the-redback.com
- linkedin: maruftuhin
- github: the-redback
- stackoverflow: 4628962
- scholar: Prun7T8AAAAJ
- codeforces: the_redback

TECHNICAL SKILLS

GO, Java, C++, TypeScript, Python, Terraform

Kubernetes, Docker, Helm, ArgoCD, Linkerd, Istio

Azure, AWS, GCP, Linux, Bash, Microservices

PostgreSQL, MySQL, MongoDB, RabbitMQ, Kafka

Prometheus, Grafana, Github Actions, Ansible, CI/CD

TOP PROJECTS

KubeDB

https://kubedb.com

Kubernetes controller powering a database management platform, automating provisioning, scaling, backup, recovery, and failover.

- Led the team to design & implement the core features.
- Designed and implemented MongoDB clustering system support.
- Improved PostgreSQL and Elasticsearch clustering system.
- Worked with Prometheus exporters to export database metrics.

Stash

https://stash.run

Backup automation tool for Kubernetes. Under the hood, it uses **Restic** for backup/restore purposes.

- Improved database backup system using stash.
- Designed and implemented Elasticsearch and MongoDB cluster backup system from scratch.
- Integrated Stash support within Kubedb to backup/restore dbs.

Microservice Architecture (MSA) Reconstruction

- Software architecture reconstruction (SAR) from tracing log.
- Automatically detection of anti-patterns from SAR.
- Used **Istio** to collect the tracing log, **Java** to process it, and Graphviz to generate the architecture graph.

SELECTED PUBLICATIONS

- "Microservice Architecture Reconstruction and Visualization Techniques: A Review." IEEE SOSE, 2022. (110 citations)
- "On Microservice Analysis and Architecture Evolution: A Systematic Mapping Study." Applied Sciences, 2021. (106 citations)
- "Catalog and Detection Techniques of Microservice Anti-Patterns and Bad Smells: A Tertiary Study." Journal of Systems and Software, 2023. (63 citations)
- "Using Microservice Telemetry Data for System Dynamic Analysis." IEEE SOSE, 2022. (56 citations)
- Full list: Google Scholar (600+ citations)